

1 Problem statement

$$\begin{aligned}
\widehat{H}\psi &= i\hbar \frac{\partial \psi}{\partial t} \\
-\frac{\hbar^2}{2m} \nabla^2 \psi + U(\vec{r}, t) \psi &= i\hbar \frac{\partial \psi}{\partial t} \\
\frac{i\hbar}{2m} \nabla^2 \psi - \frac{iU(\vec{r}, t)}{\hbar} \psi &= \frac{\partial \psi}{\partial t} \\
a \nabla^2 \psi + \widetilde{U}(\vec{r}, t) \psi &= \frac{\partial \psi}{\partial t} \\
\begin{cases} \frac{\partial \psi}{\partial t} = a \left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right) \psi + \widetilde{U}(\vec{r}, t) \psi \\ \psi|_{\Gamma} = 0 \end{cases} \\
a = \frac{i\hbar}{2m} \quad \widetilde{U}(\vec{r}, t) &= -\frac{iU(\vec{r}, t)}{\hbar}
\end{aligned}$$

2 Numerical scheme

$$\begin{aligned}
\frac{\partial \psi}{\partial t} &= a \left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right) \psi + \widetilde{U}(\vec{r}, t) \psi \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \int_{z_{k-\frac{1}{2}}}^{z_{k+\frac{1}{2}}} \cdot dx dy dz \\
\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} f(x) dx &= f_i h_x \\
\frac{\partial \psi}{\partial t} \{i, j, k\} h_x h_y h_z &= \\
a \frac{\psi\{i+1, j, k\} - \psi\{i, j, k\}}{h_x} h_y h_z - a \frac{\psi\{i, j, k\} - \psi\{i-1, j, k\}}{h_x} h_y h_z + \\
a \frac{\psi\{i, j+1, k\} - \psi\{i, j, k\}}{h_y} h_x h_z - a \frac{\psi\{i, j, k\} - \psi\{i, j-1, k\}}{h_y} h_x h_z + \\
a \frac{\psi\{i, j, k+1\} - \psi\{i, j, k\}}{h_z} h_x h_y - a \frac{\psi\{i, j, k\} - \psi\{i, j, k-1\}}{h_z} h_x h_y + \\
\widehat{U}\{i, j, k\}(t) h_x h_y h_z \psi\{i, j, k\}
\end{aligned}$$

$$\begin{aligned}
& \frac{\partial \psi}{\partial t} \{i, j, k\} = \\
& \frac{a}{h_x^2} \psi \{i+1, j, k\} + \frac{a}{h_x^2} \psi \{i-1, j, k\} + \\
& \frac{a}{h_y^2} \psi \{i, j+1, k\} + \frac{a}{h_y^2} \psi \{i, j-1, k\} + \\
& \frac{a}{h_z^2} \psi \{i, j, k+1\} + \frac{a}{h_z^2} \psi \{i, j, k-1\} + \\
& \left(-\frac{2a}{h_x^2} - \frac{2a}{h_y^2} - \frac{2a}{h_z^2} + \widehat{U} \{i, j, k\}(t) \right) \psi \{i, j, k\}
\end{aligned}$$

$$\begin{cases} \frac{\partial \Psi}{\partial t} = A(t) \Psi \\ \Psi|_{t=0} = \Psi_0 \end{cases}$$

$$A(0) \Psi_0 = \lambda_0 \Psi_0$$

$$\begin{aligned}
& \frac{\Psi_{n+1} - \Psi_n}{\Delta t} = A_n \Psi_n \\
& \Psi_{n+1} = (A_n \Delta t + I) \Psi_n \\
& \Psi_N = \left(\prod_{n=0}^{N-1} (A_n \Delta t + I) \right) \Psi_0
\end{aligned}$$